Jungbin Janice Min 민정빈



Contact Information

Address: Yangcheon-gu, Seoul, Republic of Korea

Email: janicemin2000@gmail.com

Affiliation: Sungkyunkwan University, Mechanical Engineering (Senior)

Research Interests

• Robot Design, Modeling, Control, and Simulation-Based Performance Evaluation
• Soft Wearable Robotics and Tendon-Driven Robotic Mechanisms
• Generative AI and Physics-Based Simulation for Robot Design
• Safety and Reliability Evaluation of Robotic Systems

Education

Sungkyunkwan University, School of Mechanical Engineering | 2023–Present

B.S. in Mechanical Engineering (Expected Graduation: 2027)

- GPA: 4.25 / 4.5

Research and Extracurricular Activities

Undergraduate Researcher, BEAR Research Lab (SKKU) | Aug 2026–Present

- Model and simulate tendon-driven robotic grippers using MuJoCo.
- Optimize gripper geometry and assess grasping across object sizes and positions.

Research Project: ***Before We Trust Them: Decision-Making Failures in Navigation of Foundation Models***

Carnegie Mellon University, roBot Intelligence Group (BIG) | 2025–2026

- Studied limitations of LLM-based navigation decision-making.
- Analyzed spatial-reasoning errors, uncertainty, and constraint violations.
- Evaluated LLM decisions in ambiguous and safety-critical navigation environments.
- Submitted this research to **AAAI Summer Symposium 2026**.

President, Didimdol Student Club, Sungkyunkwan University | 2025

- Led the Didimdol student club, focused on social science seminars and discussions.
- Organized regular seminars and discussions to encourage participation.
- Managed activity planning, task delegation, and communication among club members.

Tutor, Sungkyun Tutoring Program – General Physics I | Spring 2024

- Selected as a tutor based on strong academic performance in General Physics I.
- Led weekly group tutoring sessions on fundamental physics concepts.
- Supported student learning through concept explanations and problem-solving guidance.

Research Assistant, Biological Psychiatry Laboratory, Yonsei University College of Medicine | 2020

- Conducted social interaction experiments using laboratory mice.
- Used DeepLabCut to track body-part positions in mice and analyze social interaction behaviors.

Honors and Scholarships

Academic Excellence Scholarship, Sungkyunkwan University | Spring and Fall 2024

- Received a merit-based scholarship awarded to students in the top 5% for academic achievement.
- Selected based on high GPA and fulfillment of the required academic credit load.